Assignment 7 – Climate Change Tweets

In this assignment, you will analyze a dataset of climate-related tweets collected from different countries. Each row represents one tweet discussing a climate topic such as air quality, emissions, drought, renewable energy, or global warming.

Dataset File: Climate_Tweets_Sample.csv

Note: There are a few missing values and noisy entries. You must clean the data before visualization.

Part A – Cleaning & Sentiment Overview (20 marks)

- Proper date parsing (convert the date column to a proper format)
- Duplicate detection and handling.
- Handling missing values in columns (like_count/retweet_count) with proper justification.
- Clear documentation of cleaning steps.

Sentiment Visualization

- Bar chart that shows the overall distribution of tweet sentiments (e.g., Positive / Neutral / Negative) across all countries.
- Identify the top 10 countries with the most tweets, visualize the sentiment distribution across these countries.
- Proper labels, legends, titles.

Part B – Text Preprocess & Word Cloud (25 marks)

Text Preprocessing

- Lowercasing, punctuation removal, stopwords removal (common words such as "the", "is", etc.)
- Optional lemmatization / extra cleaning clearly explained

Keyword & Word Cloud Analysis

- Identify the Top 20 most frequent terms in the dataset, visualize them using a bar chart.
- Word cloud – Positive tweets.
- Word cloud – Negative tweets.
- Proper labeling & clarity.

Part C – Geospatial Visualization (15 marks)

- Correct country aggregation
- Proper choropleth implementation

- Clear color scale, legend, title
- Handling country name inconsistencies

Part D – Uncertainty & Association (25 marks)

Monthly Negative Sentiment Rate + 95% confidence interval (CI).

Compute and visualize the monthly rate of negative tweets with uncertainty.

- Correct rate calculation
- Correct CI method (Wilson or Normal)
- Proper time-series visualization with CI
- Explain/discuss your method

Scatter/Bubble: Likes vs Retweets (10 marks)

- To avoid an overly dense plot, select a subset of countries (for example the top 5 countries with the most tweets or another clearly justified selection).
- Correct scatter implementation
- Coloring by sentiment
- Insightful discussion

Part E – Design & Insight (15 marks)

- Clear explanation of design choices (color, scale, uncertainty)
- Insight paragraph (4–6 sentences, analytical not descriptive only)
- Overall clarity, readability, and professionalism

Deliverables (Required)

- Jupyter Notebook (.ipynb)
- PDF report
- Cleaned dataset
- All figures must include titles, axis labels, legends, and brief captions